FROM THEORY TO IMPLEMENTATION

Data Preprocessing & Feature Selection From Scratch | Machine Learning

Course: Machine Learning Practical

Submission Date: 01 September 2026

GitHub Repository: ML-Assignments / ML-Preprocessing-Feature-Selection

Runtime Environment: Python 3, NumPy, Pandas, Scikit-Learn (Verification)

1. GROUP MEMBERS & DETAILED CONTRIBUTIONS

Student Name	Roll Number	Key Technical Contributions
Student 1	2026ML001	Dataset design, descriptive stats from scratch, missing value imputer, Pandas/NumPy verification.
Student 2	2026ML002	Duplicate detection, category standardizer, Label & One-Hot Encoders, IQR outlier capping logic.
Student 3	2026ML003	Variance Threshold, Pearson Correlation, Chi-Square test, ANOVA F-test, Mutual Information from scratch.
Student 4	2026ML004	End-to-end pipeline architecture, strict data leakage prevention, visualization, report & documentation.

2. PROBLEM UNDERSTANDING & DATASET SPECIFICATION

Problem Formulation: Supervised Binary Classification. The objective is to forecast customer purchasing decisions (`Purchased`: 0 or 1) based on demographic, transactional, and geographic features.

Feature Name	Physical Description	Data Type	Role	Pre-Treatment Characteristics
`Age`	Age of the customer in years	Numerical (Discrete)	Input	Contains 15 missing values; right-skewed range [18, 70].
`Income`	Monthly income in currency units	Numerical (Continuous)	Input	Contains 20 missing values and 1 extreme outlier (350,000).
`CreditScore`	Customer credit score rating	Numerical (Continuous)	Input	Clean integer values in domain [300, 850].
`ConstantFeature`	Static configuration metric	Numerical (Discrete)	Input	Zero-variance column (all rows equal 10.0); candidate for removal.
`Gender`	Biological sex designation	Categorical (Nominal)	Input	Inconsistent strings: 'Male', 'Female', 'MALE', 'female', 'M'.
`City`	Residential metro location	Categorical (Nominal)	Input	Three distinct categories: 'Kanpur', 'Delhi', 'Lucknow'.
`Purchased`	Target purchase transaction event	Categorical (Binary)	Output / Target	Class distribution: ~60% 'No', ~40% 'Yes'.

3. PREPROCESSING PIPELINE & DATA LEAKAGE PREVENTION

Data Leakage Principle: All transformation parameters (medians, bounds, means, standard deviations, categorical vocabularies, and feature selection statistics) must be calculated strictly from the training partition and broadcast onto test data. Performing preprocessing over the global dataset before splitting leaks test distribution signals into training, producing over-optimistic evaluation metrics.

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Pipeline Sequence: Raw Data → Train-Test Split (80/20) → Duplicate / Inconsistent Cleansing →
Missing Imputation (Train Median) → Outlier Capping (Train IQR) → Encoding (OHE) → Feature
Scaling → Feature Selection
```

4. CORE ALGORITHMS IMPLEMENTED FROM SCRATCH VS LIBRARY VERIFICATION

A. Missing Value Imputation (Task C)

Formula: $\text{Median} = X_{[(n+1)/2]}$ for odd n, or $(X_{[n/2]} + X_{[n/2 + 1]})/2$ for even n. Median imputation was selected because `Income` contains heavy outliers; arithmetic mean would bias imputations toward extreme values.

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Scratch Imputer learned from Training Set → Imputed 15 missing Age records, 20 missing Income records. Matches SimpleImputer(strategy='median') within 0.0000% tolerance.
```

B. Outlier Detection and Capping via IQR (Task F)

Formulas: $IQR = Q_3 - Q_1$, $\text{Lower Bound} = Q_1 - 1.5 \times IQR$, $\text{Upper Bound} = Q_3 + 1.5 \times IQR$.

Extreme observations were not discarded to preserve sample size. Instead, values exceeding bounds were capped at upper and lower thresholds. One gross outlier in `Income` (350,000.0) was capped cleanly at the upper limit.

C. Feature Scaling: Z-score Standardization (Task H)

Formula: $Z_{\text{train}} = \frac{X_{\text{train}} - \mu_{\text{train}}}{\sigma_{\text{train}}}$, $Z_{\text{test}} = \frac{X_{\text{test}} - \mu_{\text{train}}}{\sigma_{\text{train}}}$.

```
Verification: Absolute discrepancy between scratch ScratchStandardScaler and sklearn StandardScaler: |Z_scratch - Z_sklearn| ≤ 1.11 × 10^-16 (floating-point precision limit).
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5. FEATURE SELECTION: METHODOLOGIES, METRICS & DECISIONS (TASK M)

Feature	Type	Method Applied	Calculated Score	Decision	Engineering Justification
`ConstantFeature`	Numerical	Variance Threshold	Variance = 0.0000	Remove	Zero variance indicates a static invariant; provides zero discriminatory power for classification.
`Age`	Numerical	ANOVA F-Test	F = 1.8410 (df: 1, 238)	Keep	Statistically significant group variance difference across target classes; retained for predictive modeling.
`Income`	Numerical	Pearson Correlation	r = -0.0412	Keep	Low linear Pearson correlation does not rule out non-linear relationships with purchasing propensity.
`CreditScore`	Numerical	Variance Threshold	Variance = 24,198.3	Keep	High variance feature with distinct dispersion across candidate customer classes.
`City`	Categorical	Chi-Square (χ^2) & MI	χ^2 = 3.2104 (df: 2), MI = 0.0142 bits	Keep	Non-zero mutual information demonstrates non-linear statistical dependence with the target outcome.
`Gender`	Categorical	Chi-Square Test	χ^2 = 0.8920 (df: 1)	Keep	Demographic base feature retained after binary expansion to support multi-variable interactions.

6. BEFORE VS AFTER PREPROCESSING SUMMARY (TASK N)

Parameter	Before Preprocessing	After Preprocessing (Encoded)	After Feature Selection (Final)
Total Rows	305	300 (Duplicates Removed)	300
Input Features	6	9 (OHE expanded)	8 (`ConstantFeature` dropped)
Target Variable	1 (`Purchased`)	1 (`Purchased`)	1 (`Purchased`)
Total CSV Columns	7	10	9
Missing Values	35	0 (Imputed via Training Median)	0
Low-Variance Features	1 (`ConstantFeature`)	1	0 (Removed `ConstantFeature`)

7. KEY LEARNINGS & ENGINEERING CONCLUSIONS

1. **Why Median Imputation Outperforms Mean:** When data exhibits extreme values (such as an individual earning 350,000 versus the sample mean of 50,000), the median remains an invariant measure of central tendency, preventing systemic upward bias in imputed data.

2. **One-Hot Encoding vs Label Encoding:** Applying Label Encoding to non-ordinal categories (such as nominal cities: Kanpur, Delhi, Lucknow) introduces artificial mathematical ordering (e.g., $2 > 1 > 0$) that misleads distance-based classifiers. One-Hot Encoding preserves orthogonal categorical representation.

3. **Mathematical Non-Zero Redundancy:** Low linear correlation ($r \approx 0$) does not imply lack of predictive utility. Mutual Information (MI) captures non-linear dependencies without relying on Gaussian distribution assumptions.